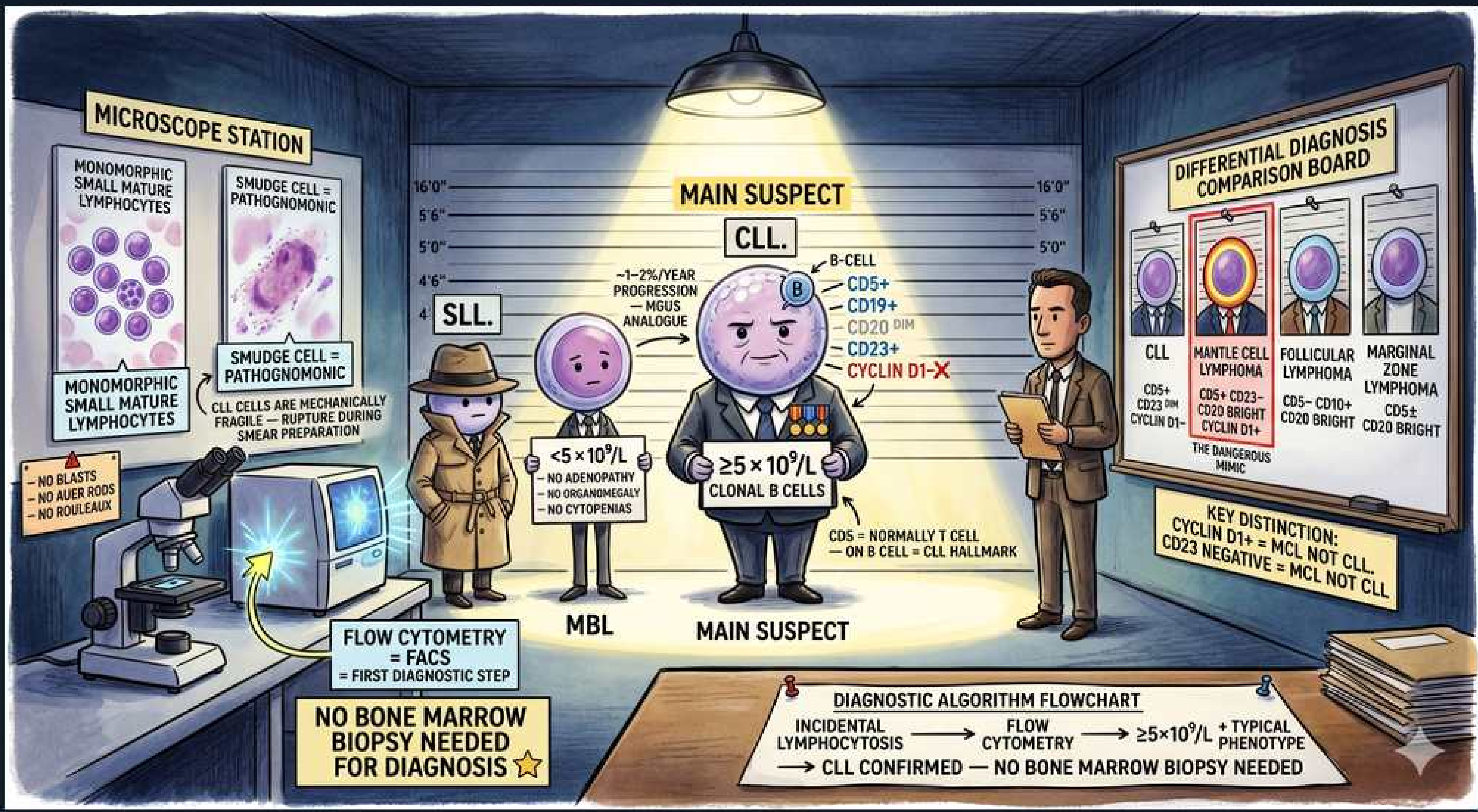
# CLL Diagnosis: Flow Cytometry, MBL & SLL



## 1. Smudge cells (pathognomonic)

**WHAT YOU SEE**

Microscope station showing smudge/basket cells and small mature lymphocytes

**WHAT IT ENCODES**

Peripheral smear shows monomorphic small mature lymphocytes plus smudge (basket) cells — fragile cells crushed during smear prep. Smudge cells are a classic CLL clue.

**BOARD TRAP**

Smudge cells are an artifact of mechanical fragility, not a true morphologic cell type; they suggest CLL but flow cytometry confirms the diagnosis.

## 2. >=5 x 10^9/L clonal B cells

**WHAT YOU SEE**

The 'Main Suspect' general with >=5 x10^9/L clonal B cells

**WHAT IT ENCODES**

CLL requires >=5,000/uL (>=5 x10^9/L) monoclonal B lymphocytes in blood for >=3 months, with the characteristic immunophenotype. This threshold separates CLL from MBL.

**BOARD TRAP**

The >=5 x10^9/L count must be CLONAL B cells, not total lymphocytes — a reactive lymphocytosis with polyclonal cells is not CLL.

## 3. CLL immunophenotype (CD5+CD19+CD23+)

**WHAT YOU SEE**

Phenotype list: CD5+, CD19+, CD23+, Cyclin D1 negative

**WHAT IT ENCODES**

CLL coexpresses CD5 (normally a T-cell marker) on B cells with CD19, CD20 (dim), CD23 positive, and dim surface immunoglobulin. CD5 on a B cell is the hallmark.

**BOARD TRAP**

CD5+ also appears in mantle cell lymphoma — the discriminators are CD23 POSITIVE and Cyclin D1 NEGATIVE in CLL (opposite in MCL).

## 4. MBL (<5 x10^9/L, no symptoms)

**WHAT YOU SEE**

Small figure labeled MBL with <5 x10^9/L, no adenopathy, no cytopenias

**WHAT IT ENCODES**

Monoclonal B-cell lymphocytosis = clonal CLL-phenotype B cells <5 x10^9/L with NO adenopathy, organomegaly, or cytopenias. Progresses to CLL at ~1-2% per year.

**BOARD TRAP**

MBL is a precursor: same phenotype but below the count threshold and asymptomatic. Don't diagnose CLL if count <5 x10^9/L and no other features.

## 5. SLL (tissue, nodal)

**WHAT YOU SEE**

Trench-coat figure labeled SLL, the tissue 'mouse analogue'

**WHAT IT ENCODES**

Small lymphocytic lymphoma is the same disease as CLL but predominantly nodal/tissue-based with <5 x10^9/L circulating clonal B cells. Diagnosed on lymph node biopsy.

**BOARD TRAP**

SLL and CLL are one entity; the distinction is WHERE the cells are (nodes/tissue in SLL vs blood in CLL), not a different biology.

## 6. Mantle cell distractor (Cyclin D1+)

**WHAT YOU SEE**

Comparison board with mantle cell lymphoma highlighted, Cyclin D1+

**WHAT IT ENCODES**

Mantle cell lymphoma is CD5+ like CLL but CD23 negative and Cyclin D1 positive (t(11;14)). This is the most dangerous CLL mimic on boards.

**BOARD TRAP**

If a CD5+ B-cell neoplasm is Cyclin D1 POSITIVE / CD23 NEGATIVE, it is mantle cell — NOT CLL. The opposite pattern confirms CLL.

## 7. Follicular lymphoma distractor

**WHAT YOU SEE**

Comparison panel: follicular lymphoma CD5- CD10+

**WHAT IT ENCODES**

Follicular lymphoma is CD5 negative, CD10 positive, with t(14;18)/BCL2. Distinguishes from CLL which is CD5+ CD10-.

**BOARD TRAP**

CD10 positivity points to follicular (germinal-center origin), not CLL; CLL is CD5+ CD10-.

## 8. Marginal zone distractor

**WHAT YOU SEE**

Comparison panel: marginal zone lymphoma CD5- CD20 bright

**WHAT IT ENCODES**

Marginal zone lymphoma is CD5 negative with bright CD20. Differentiated from CLL's CD5+ and dim CD20.

**BOARD TRAP**

Bright CD20 with CD5 negativity favors marginal zone; CLL classically has DIM CD20 and is CD5 positive.

## 9. Flow cytometry = first step

**WHAT YOU SEE**

Flow cytometry / FACS machine labeled 'first diagnostic step'

**WHAT IT ENCODES**

Peripheral blood flow cytometry establishes clonality and immunophenotype and is the single most important first diagnostic test in suspected CLL.

**BOARD TRAP**

Flow cytometry on PERIPHERAL BLOOD is diagnostic — you do not need a bone marrow biopsy or lymph node biopsy to diagnose CLL.

### 10. No bone marrow biopsy needed

WHAT YOU SEE

Banner: 'no bone marrow biopsy needed for diagnosis'

WHAT IT ENCODES

CLL diagnosis is made on peripheral blood flow cytometry alone when clonal B cells >=5 x10^9/L with typical phenotype. Marrow biopsy is reserved for unexplained cytopenias.

BOARD TRAP

Ordering a bone marrow biopsy to confirm CLL is unnecessary and a classic wrong answer — blood flow cytometry suffices.

---

### 11. Diagnostic algorithm flowchart

WHAT YOU SEE

Bottom flowchart: incidental lymphocytosis -> flow -> >=5 x10^9/L + typical phenotype -> CLL confirmed

WHAT IT ENCODES

Workflow: incidental lymphocytosis -> peripheral flow cytometry -> if >=5 x10^9/L clonal B cells with typical phenotype -> CLL confirmed, no marrow needed.

BOARD TRAP

If clonal count is <5 x10^9/L the answer is MBL, not CLL — the algorithm hinges on the absolute clonal B-cell count.

---